Supplementary Information for:

Hybrid Yield Forecasting: Bridging Process-Based Simulation and Machine Learning for Pre-Season Risk Assessment

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Abstract

This document provides technical details, comprehensive data specifications, model hyperparameters, and regional forensic analyses supporting the main text. This material ensures the reproducibility of the Super Ensemble pipeline and provides evidence for the regime-switching hypothesis.

Contents

S1 Master Data Table

The operational forecasting pipeline integrates 11 distinct data streams to bridge the gap between seasonal climate precursors and field-level physiological responses.

Table S1: Comprehensive Data Sources and Specifications

Dataset	Source	Resolution	Role
Yield Data	Thuringian State Inst. / Destatis [Statistisches Bundesamt (Destatis), 2024]	District (Annual)	Target Variable
AgERA5	ECMWF / Copernicus [Service, 2024]	0.1° (Daily)	Historical Weather
SEAS5	ECMWF [Johnson et al., 2019]	1.0° (Monthly)	Seasonal Forecast
ERA5-Land	ECMWF [Muñoz-Sabater et al., 2021]	0.1° (Monthly)	Soil Moisture / Temp
SoilGrids	ISRIC [Hengl et al., 2017]	250m	Static Soil Properties
CropMap	DLR [DLR (German Aerospace Center), 2023]	10m	Spatial Mask
MOD13Q1	NASA LP DAAC [Didan, 2015]	250m (16-day)	Vegetation Anomalies

S2 Extended Mechanistic Formulas

The **Compound Failure Index** identifies yield crashes by aggregating stress vectors derived from WOFOST internal moisture balances [Boogaard et al., 1998] and SEAS5 anomalies.

Scorch Index (Heat + Drought interaction):

$$\text{Scorch} = \max\left(Z_{\text{heat}} \times (-Z_{\text{bal}})^{+}, (Z_{\text{heat}} - 1.5)^{+} \times 2.0\right) \tag{S1}$$

Drown Index (Anoxia/Waterlogging):

$$\text{Drown} = (Z_{\text{anoxia}} - 0.8)^{+} \times 2.0 \tag{S2}$$

Late Start Index (Sowing delay):

$$\text{LateStart} = (Z_{\text{sow}} - 1.5)^{+} \times 2.0 \tag{S3}$$

S3 Comprehensive Hyperparameters

Hyperparameters for the Meta-Learner were configured following best practices for robust generalization in high-noise agricultural environments [Paudel et al., 2022].

Table S2: Hyperparameters for Sub-Models and Meta-Learner

Model	Configuration
M_1 Trend	LinearGAM ($n\_splines = 10$), ARIMA$(1, 0, 0)$
M_2 Physics V2	XGBoost: $n\_est = 2000, lr = 0.01, max\_depth = 7$
M_3 Hybrid	XGBoost: $n\_est = 1242, lr = 0.057, max\_depth = 3$ [Shahhosseini et al., 2021]
M_4 Robust	HuberRegressor: $\epsilon = 1.35, max\_iter = 200$
Meta-Learner	XGBoost: $n\_est = 200, lr = 0.03, max\_depth = 4$

S4 Implementation Details

As discussed in Section 2.6 of the main text, the M_3 component targets the yield-to-trend ratio rather than absolute yield to improve local optimization convergence[cite: 421, 422].

Yield-to-Ratio Scaling (M_3):

```
# Predicting the Ratio between observed yield and WOFOST simulation
df['yield_ratio'] = df['actual_yield'] / df['stage1_forecast']
df['yield_ratio'] = df['yield_ratio'].clip(0.5, 1.5)
```

References

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